Force types

- Forces acting on a system can be divided into two types according to how they affect potential energy.
- *Conservative forces* can be related to potential energy changes.
- *Non-conservative forces* cannot be related to potential energy changes.
- So, how exactly do we distinguish between these two types of forces?

Conservative forces

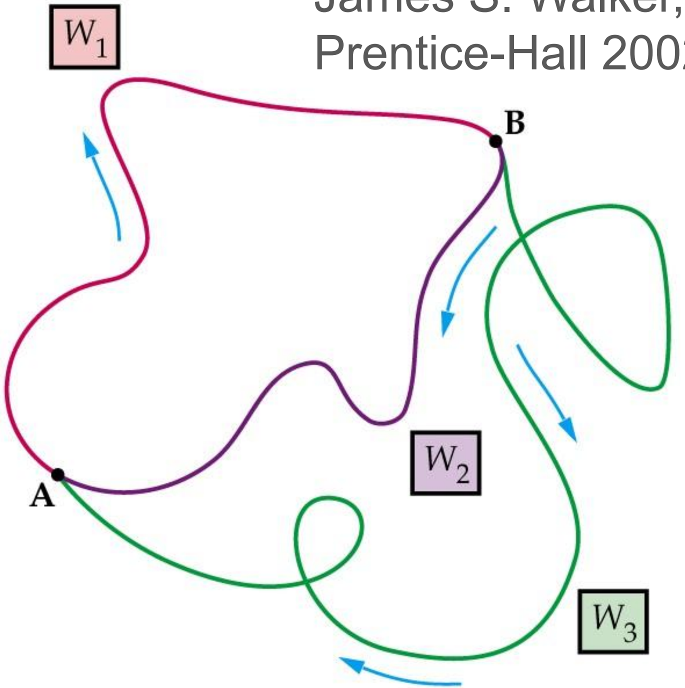
- Work is ***path independent***.
  - Work can be calculated from the starting and ending points only.
  - The actual path is ignored in calculations.
- ***Work along a closed path is zero***.
  - If the starting and ending points are the same, no work is done by the force.
- ***Work changes potential energy***.
- Examples:
  - Gravity
  - Spring force

Conservative forces and Potential energy

- $W_c = -\Delta U$
- If a conservative force does positive work on a system, potential energy is lost.
- If a conservative force does negative work, potential energy is gained.
- For gravity
  - $W_g = -\Delta U_g = -(mgh_f - mgh_i)$
- For springs
  - $W_s = -\Delta U_s = -(\frac{1}{2} k x_f^2 - \frac{1}{2} k x_i^2)$

More on paths and conservative forces.

Figure from "Physics", James S. Walker, Prentice-Hall 2002



Q: Assume a conservative force moves an object along the various paths. Which two works are equal?

A: $W_2 = W_3$

(path independence)

Q: Which two works, when added together, give a sum of zero?

A: $W_1 + W_2 = 0$

or

$W_1 + W_3 = 0$

(work along a closed path is zero)

Non-conservative forces

- Work is *path dependent*.
  - Knowing the starting and ending points is not sufficient to calculate the work.
- *Work along a closed path is NOT zero*.
- *Work changes mechanical energy*.
- Examples:
  - Friction
  - Drag (air resistance)
- *Conservation of mechanical energy does not hold!*